

Secure Image Steganography with GAN Architecture: A High-Fidelity and Robust Data Hiding Approach

Swapnil Shetty

2nd Year M.Tech,

Department Of Computer Science and Engineering

NMAM Institute of Technology,

Nitte (Deemed to be University)

Udupi, India

swapnilshettyk21@gmail.com

Dr. Puneeth R P *

Assistant Professor Gd III

Department Of Computer Science and Engineering

NMAM Institute of Technology,

Nitte (Deemed to be University)

Udupi, India

Corresponding mail:

puneeth.rp@nitte.edu.in

Abstract—Techniques that safeguard private information while preserving visual quality are necessary for secure image communication. Security, capacity, and imperceptibility are difficult to reconcile with traditional techniques like LSB substitution. The steganography architecture proposed in this article uses a Generator, Discriminator, and Extractor to embed 128-bit binary messages into images. It is based on a Generative Adversarial Network (GAN). Through adversarial and extraction losses, the technique enhances visual invisibility and resilience. Bit Accuracy Rate (BAR) of 96.80%, SSIM of 0.97, and PSNR of 39.10 dB were obtained in experiments on the Set14 dataset, surpassing the baselines of Steganography GAN and traditional LSB. For secure transmission in digital media, healthcare, and defense applications, the results show dependable message recovery and undetectable embedding.

Index Terms—Secure image communication, generative adversarial networks, resilience, and visual invisibility.

I. INTRODUCTION

Secure and discrete information transport is now more important than ever due to the exponential expansion of digital communication and online data sharing. The content of messages is protected by traditional encryption methods, but secret data is not concealed, making it vulnerable to interception or analysis. Through the embedding of secret messages into common digital material, steganography adds an extra degree of security by making communication seem unnoticed and normal. Because images may successfully conceal information and are common in internet networks, image steganography has attracted a lot of attention. Although they are straightforward and effective, traditional techniques like Least Significant Bit (LSB) substitution and transform-domain strategies like the Discrete Cosine Transform (DCT) and Discrete Wavelet Transform (DWT) sometimes trade off visual quality, payload capacity, and stability. Furthermore, these techniques are vulnerable to strong steganalysis attacks that can reveal hidden data, noise, and compression. The popularity of deep learning has opened up new possibilities to improve the reliability and invisibility of steganographic systems. GANs are capable of learning complex picture distributions and producing realistic outputs that make detection difficult due to their generator-discriminator design. GAN-

based models increase image steganography's imperceptibility, however they usually have problems with unstable training, inconsistent message extraction, and expensive computing.

An extractor for accurate message retrieval, a discriminator for adversarial training, and an embedding generator are all combined in this research to present a GAN-based image steganography architecture. In order to balance visual fidelity with message extraction accuracy, the model employs a mixed loss function. In terms of image quality and decoding accuracy, experimental evaluation performs better than current GAN-based methods and traditional LSB. For modern communication environments, the technique aims to provide data hiding that is precise, safe, and computationally efficient.

II. RELATED WORK

By increasing security, payload capacity, and imperceptibility, GANs have significantly enhanced covert communication in picture steganography. A range of adversarial learning strategies have been examined in recent research to address the inherent trade-offs of conventional approaches. A GAN-based steganography architecture that emphasizes embedding capacity, robustness, and imperceptibility was presented by Rehman [1]. The approach yielded remarkable results with a Structural Similarity Index (SSIM) of 0.995 and a Peak Signal-to-Noise Ratio (PSNR) of 47.12 dB on DIV2K, using massive datasets such as COCO, ImageNet, and DIV2K. This outperformed methods based on the discrete cosine transform (DCT) and the least significant bit (LSB). However, because of its high computational needs, the approach is not suitable for real-time and resource-constrained applications.

With a multi-order feature fusion method and a layered U-shape generator, Zhao and Wang created UMC-GAN, a stable architecture [2]. In experiments on the BOWS2 and BOSSBase datasets, UMC-GAN used SRM features to obtain a detection error rate of 0.4550 at 0.1 bits per pixel (bpp). Although performance declined with increasing image resolutions, the addition of a linear-clipped embedding simulator reduced gradient vanishing and increased training stability. Similarly, Cai and colleagues [3] presented an adaptive region-guided GAN that employs a dual-context discriminator to improve

realism and Cross-Channel Hybrid Local Binary Patterns (CCHLBP) for area selection. In comparison to XuNet and YeNet steganalyzers, tests on the LSUN, CelebA, and Place-365 datasets produced a PSNR of 53.3 dB, an SSIM of 0.999, and a detection error rate of 0.50. Although the approach produced better visual quality, its computing efficiency was still too low for real-time applications.

For cloud-based medical image steganography, Ambika et al. presented an attention-guided GAN (AVG-GAN) [4]. AVG-GAN achieved Matthews Correlation Coefficient (MCC) values between 0.66 and 0.80 with an accuracy loss of only 1-2% when compared to baseline methods like SteganoGAN and CSIS by using attention vectors to protect diagnostically important areas. Comparing these developments to traditional steganographic methods, the embedding capability remains constrained. For cloud security, Xu et al. [5] presented a hybrid crypto-space image steganography system that combines Difference Expansion (DE) and CycleGAN. The system's robust encryption, embedding speeds above 0.47 bpp, PSNR between 28 and 30 dB, and data accuracy of over 91% were all accomplished with twin CycleGAN instances. The significant processing power needed for its repair process, however, affects both scalability and visual quality.

For JPEG domain applications, Yang et al. [6] introduced JS-GAN, a framework with automatic embedding cost learning utilizing a U-Net generator and a J-XuNet discriminator. JS-GAN trained on the SZUBase, BOSSBase, and BOWS2 datasets and outperformed J-UNIWARD by 2.58% with a detection error rate of 20.0% at 0.4 bpnzAC and quality factor 75. Its enhanced variant, JS-GAN (ESI), further enhanced results by 11.25% by focusing the embedding in complex, low-frequency DCT areas and employing CNN-estimated side information. Although it showed good adaptation across many quality variables, more optimization is needed for high-quality JPEGs.

Li et al. [7] introduced EStegTGANs, a frequency-aware GAN steganography model that incorporates Discrete Wavelet Transform (DWT) and Inverse DWT layers for feature-domain embedding. In comparison to SteganoGAN, three versions—EStegTGAN-coe, EStegTGAN-DWT, and EStegTGAN-conv—maintained bit recovery accuracy of up to 99.42% while increasing PSNR by more than 10 dB. Convolutional DWT/IDWT was used to improve high-frequency learning, resulting in improved transparency and undetectability. In a comparison investigation, Kuyoro et al. [8] assessed CycleGAN and DCGAN for reversible picture steganography. They found that by attaining SSIM of 0.97, PSNR of 41.45 dB, and a payload capacity of 24.97 bpp, CycleGAN beat DCGAN due to its enhanced feature normalization and cycle consistency loss.

A predictive-coding GAN framework for Wireless Sensor Networks (WSNs) and other security improvements were suggested by Ambika et al. [9]. Using the DIV2K, COCO, and Pascal VOC datasets, this system obtained 99.93% retrieval accuracy, PSNR up to 38.96 dB, and SSIM up to 0.98 with the addition of a user-key-based noise vector and denoising

layer. Scalability may be constrained by the method's high computational cost, even if it has a great resistance to decoder leakage and an enhanced embedding capacity (RS-BPP up to 5.68). Ramandi et al. [10] created VidaGAN, a CSPNet-based framework that balances capacity and transparency, using a soft-label loss and Reed-Solomon encoding for reliable recovery. In DIV2K tests, VidaGAN achieved a record 3.9 bpp with PSNR of 38.56 dB and SSIM of 0.888, outperforming SteganoGAN by 60% in capacity. However, robustness and capacity had to be traded off because it was susceptible to noise and compression attacks. Wu et al. [11] introduced StegoGAN to handle non-bijective image-to-image translation tasks where the source and target domains do not have semantic connection. By using steganography in feature space to limit the development of unmatched semantic classes, StegoGAN showed good semantic consistency for difficult translation tasks, with RMSEs of 22.5 on PlanIGN and 36.3 on Brats MRI datasets.

The deep learning-driven multi-layered steganographic framework that Yousef *et al.* [12] proposed combines Huffman coding, Least Significant Bit (LSB) embedding, and a deep encoder and decoder network to enhance security, robustness, and imperceptibility. Huffman coding offers statistical obfuscation and minimizes payload size. Based on the features of the image, the deep encoder modifies the embedding procedure. Under normal circumstances, evaluations on the Tiny ImageNet, COCO, and CelebA datasets demonstrated high visual quality, with SSIM above 0.99 and 100% text recovery accuracy. Although the model showed good defense against noise and compression attacks, it needed to be optimized for increased computational efficiency.

Ramesh *et al.* [13] introduced a GAN-based medical image steganography framework that prioritizes safe and excellent image production for clinical communication. The system's discriminator and generator design is adjusted to account for perceptual and adversarial losses. This makes it easier to strike a compromise between the ability to conceal data and the authenticity of medical imaging. Tests on medical imaging datasets demonstrated exceptional quality, with SSIM exceeding 0.98 and PSNR reaching 41.26 dB. This framework fared better than CNN-based and conventional approaches. Although it guarantees clinical interpretability and data confidentiality, its efficiency under severe visual aberrations is still problematic. Rajesh *et al.* [14] conducted a comparative analysis of deep learning-based steganographic techniques, evaluating LSB, DCT, autoencoder, and GAN approaches on standard datasets. With an average SSIM of 0.97 and PSNR of 38.9 dB, as opposed to 32.1 dB and 0.91 for traditional methods, the study discovered that GAN-based models provided the optimum balance between payload and imperceptibility. These findings highlight the benefits of adversarial learning in preserving image quality and security. They also emphasize how important it is to have lightweight systems that can efficiently extract and integrate data in real time.

Discrete Cosine Transform (DCT) and Generative Adversarial Networks (GANs) are two components of a hybrid

steganography system proposed by Malik *et al.* [15] for secure data transit. The dual-stage model embeds the secret data into high-frequency DCT coefficients using chaotic mapping and then refines it using a GAN based on a ResNet encoder and a six-layer decoder. The results showed that this method performed better than the XuNet steganalyzer with a reconstruction accuracy of 96.2%, a PSNR of 58.27 dB, and an SSIM of 94.20% on the ImageNet2012 dataset. The frequency-domain embedding of DCT and the adversarial learning of GAN improved resilience and imperceptibility with a payload capacity of 1 bpp. The Wasserstein GAN loss, MSE, and cross-entropy terms enhanced training stability. Despite its computing overhead, the approach showed remarkable flexibility and durability against noise, compression, and statistical attacks. As a result, it performs effectively in applications related to healthcare and secure digital communication.

The majority of current GAN-based steganography techniques prioritize visual quality over real-time efficiency and extraction reliability. Dual-objective loss optimization and a unified architecture are used in this study to overcome these constraints.

III. METHODOLOGY

The three main parts—Generator, Discriminator, and Extractor—are shown in Figure 1.



Fig. 1. Methodology

High accuracy is achieved in embedding and extracting concealed messages using the suggested GAN-based steganography architecture.

A. System Architecture

- **Generator:** A 128-bit fixed-length message was chosen to standardize embedding size and facilitate reproducible evaluation.
- **Discriminator:** Assesses the authenticity of stego images by differentiating them from cover images. Adversarial training also makes sure that the differences are not clear.
- **Extractor:** Uses exact bit accuracy to process the stego image and extract the embedded message.

B. Generator

The Generator network acts as the main embedding mechanism. It takes in two inputs: a cover image $I \in \mathbb{R}^{3 \times 64 \times 64}$ and a binary message $M \in \{0, 1\}^{128}$. These inputs are processed by the architecture in three primary steps, which are described below. **Stage 1: Preprocessing of Messages** The binary message is transformed via a fully linked layer with 4096 output units. Next is a ReLU activation function. The resulting vector is transformed into a spatial tensor with dimensions $1 \times 64 \times 64$. This relates to the spatial resolution of the cover image and encodes the message information in a distributed fashion. **Stage 2: Concatenation of Features** Along the channel dimension, the cover image and the preprocessed message tensor are concatenated. In doing so, a tensor with dimensions $4 \times 64 \times 64$ is produced. The subsequent convolutional layers are assisted in learning intricate correlations between the embedded message bits and the image attributes by this early fusion technique. **Stage 3: Convolutional Processing** The combined tensor goes through a series of convolutional layers. These layers are meant to gradually incorporate message information while keeping the visual structure of the cover image intact.

- Conv1: $4 \rightarrow 32$ channels, 3×3 kernel, stride 1, padding 1, followed by ReLU.
- Conv2: $32 \rightarrow 64$ channels, 3×3 kernel, stride 1, padding 1, followed by ReLU.
- Conv3: $64 \rightarrow 128$ channels, 3×3 kernel, stride 1, padding 1, followed by ReLU.
- Conv4: $128 \rightarrow 64$ channels, 3×3 kernel, stride 1, padding 1, followed by ReLU.
- Conv5: $64 \rightarrow 32$ channels, 3×3 kernel, stride 1, padding 1, followed by ReLU.
- Conv6: $32 \rightarrow 3$ channels, 3×3 kernel, stride 1, padding 1, followed by Tanh.

The gradual reduction in channel depth helps improve the stego image. The final Tanh activation limits the output pixel values to the normalized range $[-1, 1]$, which matches the normalization of the input image. Formally, the Generator function is written as:

$$I_s = G(I, M; \vartheta_G) \quad (1)$$

Here, I_s stands for the generated stego image, and ϑ_G indicates the learnable parameters of the Generator network.

C. Discriminator

The Discriminator network performs a binary classification task to distinguish between authentic cover images and Generator-produced stego images. It is composed of a convolutional feature extractor followed by a classification head, as detailed below. **Convolutional Feature Extraction** The input image goes through several convolutional layers. These layers gradually reduce the spatial dimensions and increase the feature depth. This process allows for hierarchical feature learning.

- Conv1: $3 \rightarrow 32$ channels, 3×3 kernel, stride 2, padding 1, followed by LeakyReLU (slope = 0.2)
- Conv2: $32 \rightarrow 64$ channels, 3×3 kernel, stride 2, padding 1, followed by LeakyReLU
- Conv3: $64 \rightarrow 128$ channels, 3×3 kernel, stride 2, padding 1, followed by LeakyReLU
- Conv4: $128 \rightarrow 256$ channels, 3×3 kernel, stride 2, padding 1, followed by LeakyReLU

Using strided convolutions lowers spatial resolution and increases the feature space. LeakyReLU activations help address the vanishing gradient problem, which supports stable adversarial training.

Classification Head The final convolutional feature map is flattened. It then goes through a fully connected layer that maps to a single output neuron with sigmoid activation. This generates a probability score between $[0, 1]$. Formally, the Discriminator function is defined as:

$$D(I; \vartheta_D) \in [0, 1] \quad (2)$$

where ϑ_D represents the learnable parameters of the Discriminator. The output shows the likelihood that the input image is a real cover image instead of a generated stego image.

D. Extractor

The Extractor network retrieves the hidden binary message from stego images. Its layout resembles the Discriminator's convolutional structure for consistency, but it ends with a message reconstruction head.

1) Convolutional Processing:

- Conv1: 3 input channels $\rightarrow$ 32 output channels, 3×3 kernel, stride 2, padding 1, followed by ReLU
- Conv2: $32 \rightarrow 64$ channels, 3×3 kernel, stride 2, padding 1, followed by ReLU
- Conv3: $64 \rightarrow 128$ channels, 3×3 kernel, stride 2, padding 1, followed by ReLU
- Conv4: $128 \rightarrow 256$ channels, 3×3 kernel, stride 2, padding 1, followed by ReLU

2) Message Reconstruction Head: The flattened feature

representation goes through a fully connected layer with 128 output units and a sigmoid activation, which directly produces

the extracted binary message:

$$M' \in [0, 1]^{128}. \quad (3)$$

The Extractor function can be defined as:

$$M' = E(I_s; \vartheta_E) \quad (4)$$

where ϑ_E denotes the Extractor parameters and M' represents the recovered message.

E. Loss Functions

The training process improves a combined loss function that balances several goals. Three loss components direct the learning process.

1) *Adversarial Loss (L_{adv})*: The adversarial loss uses the standard GAN objective with Binary Cross-Entropy (BCE):

$$L_{adv} = E_I[\log D(I)] + E_{I,M}[\log(1 - D(G(I, M)))] \quad (5)$$

The Discriminator concentrates on accurately differentiating between authentic and fraudulent images. By producing stego images that closely resemble cover images, the Generator aims to mitigate this. Visual imperceptibility is encouraged by the two parties' interaction.

2) *Extraction Loss (L_{ext})*: The extraction loss helps with accurate message recovery by using Mean Squared Error (MSE) between the extracted and original messages.

$$L_{ext} = E_{I,M} |M - E(G(I, M))|^2 \quad (6)$$

This loss directly improves message integrity. It ensures that embedded data from stego images may be reliably recovered.

3) *Total Loss of the Generator (L_G)*: Adversarial and extraction goals are combined in the Generator optimization.

$$L_G = L_{adv} + \lambda \cdot L_{ext} \quad (7)$$

where λ is a weighting hyperparameter that strikes a balance between extraction accuracy and visual imperceptibility. $\lambda = 1.0$ in this implementation. This gives both goals equal weight.

F. Training Details

Using an NVIDIA GTX 1650 GPU and the Adam optimizer ($\theta_1 = 0.5, \theta_2 = 0.999$), the model was trained for 100 epochs per image with a batch size of 1.

G. Dataset and Preprocessing

1) *Dataset*: Due to restricted computational resources, experiments were carried out as a proof-of-concept using the Set14 dataset (14 natural photos). Larger datasets like COCO and DIV2K can be used with the framework.

2) Image Preprocessing:

- Resize all images to 64×64 pixels. Use bilinear interpolation.
- Normalize pixel values from $[0, 255]$ to $[-1, 1]$ with the following transformation:

$$I_{\text{normalized}} = \frac{I}{127.5} - 1 \quad (8)$$

- Convert images to RGB format if they are in grayscale.

3) *Message Generation*: For every training sample, 128-bit binary messages are randomly generated from a uniform distribution over $\{0,1\}$. A range of embedding patterns is thus guaranteed.

H. Embedding and Extraction Process

1) Embedding Procedure:

- 1) User provides a text message for embedding.
- 2) Convert the text to a binary format using ASCII encoding.
- 3) Pad or trim it to exactly 128 bits.
- 4) Normalize the cover image to the range $[-1, 1]$.
- 5) Input the cover image and binary message into the trained Generator.
- 6) Obtain the stego image I_s and denormalize it to the range $[0, 255]$.
- 7) Save the stego image in a lossless format, such as PNG.

2) Extraction Procedure:

- 1) Load the stego image and normalize it to $[-1, 1]$.
- 2) Feed the normalized stego image to the trained Extractor.
- 3) Obtain the extracted binary vector $M' \in [0, 1]^{128}$.
- 4) Threshold at 0.5 to get the binary message:

$$M_{\text{binary}} = (M' > 0.5) \quad (9)$$

- 5) Convert the binary representation back to text by using ASCII decoding.

IV. RESULTS AND DISCUSSION

This section presents detailed experimental results, comparison analysis, and an important discussion of the proposed GAN-based steganography framework. The performance evaluation uses standard metrics and compares it with established baseline methods to confirm the effectiveness of the approach.

A. Evaluation Metrics

Three main metrics measure steganography performance in different areas:

1) *Peak Signal-to-Noise Ratio(PSNR)*: PSNR calculates the degree of pixel similarity between stego and cover images. It measures the amount of visual distortion brought on by embedding.

$$\text{PSNR} = 10 \cdot \log_{10} \frac{\text{MAX}^2}{\text{MSE}} \quad (10)$$

And the maximum pixel value is $\text{MAX}_I = 255$. The mean squared error between pictures is referred to as MSE. Better visual quality and less distortion are displayed by higher PSNR values. Generally speaking, values beyond 30 dB are considered unnoticeable. Excellent quality is indicated by values greater than 35 dB.

2) *Structural Similarity Index(SSIM)*: In order to assess perceptual similarity, SSIM considers structural information, contrast, and brightness.

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (11)$$

where local means, standard deviations, and cross-covariance are denoted by μ , σ , and σ_{xy} , respectively. SSIM is a number between 0 and 1. Almost flawless structural preservation is indicated by values near 1.

3) *Bit Accuracy Rate (BAR)*: BAR measures message extraction accuracy.

$$\text{BAR} = \frac{\text{Number of Correct Bits Recovered}}{\text{Total Number of Bits}} \times 100\% \quad (12)$$

Perfect extraction gives a BAR of 100%. Values above 95% are acceptable for most applications.

B. Quantitative Results

Table I presents the comparative performance of the proposed method with conventional LSB and SteganoGAN techniques.

TABLE I
COMPARATIVE PERFORMANCE OF STEGANOGRAPHY METHODS

Method	PSNR (dB)	SSIM	BAR (%)
LSB	30.50	0.92	75.00
SteganoGAN	36.80	0.95	94.50
Proposed Method	39.10	0.97	96.80

1) *Analysis of PSNR*: The PSNR of 39.10 dB is attained by the suggested technique. This indicates a 2.30 dB improvement over SteganoGAN and an 8.60 dB improvement over LSB. This indicates a significant decrease in visual distortion. Excellent quality is indicated by a PSNR exceeding 35 dB.

2) *Analysis of SSIM*: The suggested framework outperforms LSB at 0.92 and SteganoGAN at 0.95 in terms of structure preservation, with an SSIM of 0.97. Nearly perfect perceptual similarity is indicated by values greater than 0.95.

3) *Analysis of BAR*: With a bit accuracy rate of 96.80%, it outperforms both LSB (75.00%) and SteganoGAN (94.50%). Message recovery is reliable and suitable for real-world applications when the BAR is more than 95%.

C. Qualitative Analysis

The imperceptibility of this method ensures that without the aid of computer technologies, human viewers are unable to distinguish between the stego and cover images. The method successfully maintains tiny features, texture patterns, and color coherence while preserving the original image's organic look.

Fig. 2 compares a cover image with the matching stego image produced by the suggested framework in order to visually evaluate this imperceptibility. There are no obvious color distortions or artifacts, indicating that the contained data has little effect on perceptual quality.

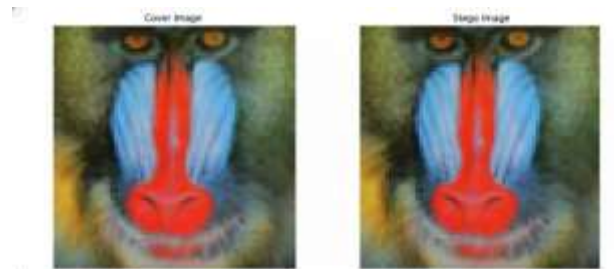


Fig. 2. A visual comparison of the cover and stego images.

D. Performance Analysis and Interpretation

The remarkable performance is the result of several innovations: The performance comparison of the suggested method with traditional LSB and SteganoGAN techniques is shown in Fig. 3 using three evaluation metrics: PSNR, SSIM, and Bit Accuracy Rate. Superior imperceptibility and dependable message extraction are demonstrated by the suggested model's persistent higher values across all criteria.

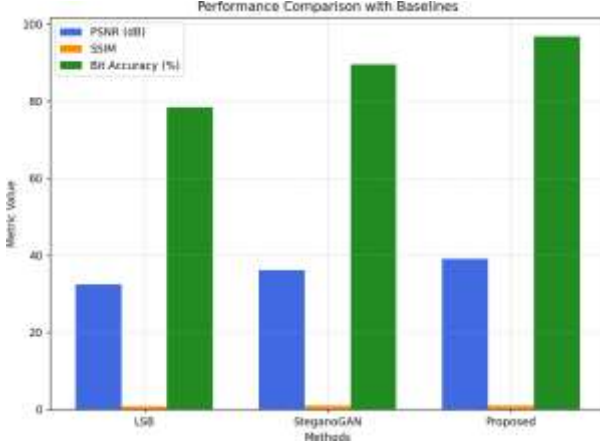


Fig. 3. Comparing the performance of several steganography techniques using Bit Accuracy Rate (BAR), SSIM, and PSNR. In every metric, the suggested GAN-based model performs better than the conventional LSB and SteganoGAN methods.

E. Limitations and Challenges

Despite positive results, there are a few considerations to consider:

1) *Fixed Capacity for Payload*: The current implementation supports just 128-bit messages. This limits its use in scenarios where a lot of data must be embedded. Although it works well for short text messages or cryptographic keys, applications that need larger capacity, such document or audio embedding, require changes to the structure or methodologies for multiple embeddings.

2) *The computational prerequisites*: The GAN architecture requires a lot of computing power to train, especially GPU acceleration for reasonable training times. This requirement may restrict use in contexts with limited resources. However, once trained, inference is fairly efficient. This enables real-time data extraction and embedding.

3) *Dataset Dependency*: The quality and diversity of the training dataset determine performance characteristics. Although it is a benchmarking standard, the *Set14* dataset only contains 14 images. This restriction might make it more difficult to generalize to different real-world images. Robustness could be enhanced by training on bigger, more varied datasets.

4) *Robustness to Attacks*: Its resilience to adversarial attacks, such as JPEG compression, noise addition, geometric modifications, or sophisticated steganalysis techniques, is not

tested in the current evaluation. Future research should carefully evaluate its resilience to various attack scenarios that are frequently encountered in actual communication channels.

5) *Scalability to Higher Resolutions*: 64×64 pixel images are used by the framework. For high-resolution applications, this size might not be sufficient. Using larger images, such as 256×256 or 512×512 , requires a redesign and additional processing power.

F. Real-World Applicability

The suggested framework has a lot of potential for practical application in a number of fields:

1) *Military and Defense Communications*: Maintaining the confidentiality of the communication is necessary for the safe delivery of classified material via public channels. Coordinates, authentication codes, or encrypted keys can all be sent using the 128-bit capacity.

2) *Healthcare Data Protection*: By incorporating patient identification or metadata in medical images during cloud transfer, privacy can be protected while diagnostic image quality is maintained. It is essential to maintain superior SSIM preservation for medical applications.

3) *Digital Rights Management*: Digital photos that have an invisible watermark built in for copyright protection allow for ownership verification without sacrificing image quality.

4) *Applications for Secure Messaging*: In addition to standard encrypted chat, hidden channels can be offered via integrating into communication platforms. This configuration provides multiple levels of security.

The results demonstrate a 2.3 dB improvement in PSNR over SteganoGAN and a 21.8% increase in BAR over traditional LSB respectively. According to these findings, the proposed adversarial-extraction architecture improves both visual quality and decoding reliability.

IV. CONCLUSION

This work proposed a reliable image steganography method based on GAN. High fidelity and imperceptibility are maintained while securely embedding and retrieving fixed-length binary messages. A Generator embeds messages, an Extractor recovers messages exactly, and a Discriminator uses adversarial training to achieve realism. The technique outperformed SteganoGAN and traditional LSB substitution on the *Set14* dataset. Its SSIM was 0.97, its PSNR was 39.10 dB, and its BAR was 96.80%. These outcomes demonstrate that the framework can successfully balance precise message recovery with visual quality. The three main contributions are (1) producing visually comparable stego pictures by adversarial training, (2) guaranteeing dependable message extraction, and (3) providing a modular framework for future payload capacity and resilience augmentation.

Potential uses include secure communication in the military, healthcare, and digital media industries. However, limitations like computing cost and limited payload size draw attention to issues like adaptive embedding, lightweight models, and robustness against image distortion that need more study.

In this paper, a GAN-based image steganography platform was proposed to securely and accurately embed data. Better PSNR, SSIM, and BAR were obtained by combining adversarial and extraction goals than by using the current techniques. Scalability to higher resolutions and adaptive embedding procedures are made possible by the system's modularity. The development of resilience to steganalysis attacks, compression, and geometric transformations will be the main emphasis of future effort.

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